Won Geun Shin

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EDUCATION

Korea University *March 2024 - Present*

Ph.D. Candidate | Advisor: Prof. HeeSeok Kim | South Korea

GPA: 4.3/4.5

Korea University *March 2018 - February 2024*

Bachelor's degree | South Korea

GPA: 4.0/4.5

RESEARCH INTERESTS

Cryptographic implementation security: side-channel analysis, fault attacks on post-quantum signatures (ML-DSA and HAETAE), masking, and secure neural-network implementations.

PUBLICATIONS AND PREPRINTS

I = International; D = Domestic; J = Journal; C = Conference; P = Preprint

[I.P.1] **Won Geun Shin**, SeungHyeon Jeon, Daehyeon Bae, Sujin Park, HeeSeok Kim. (2026). Public Coefficient Matters: A Practical Differential Fault Attack on ML-DSA and HAETAE. *IACR Cryptology ePrint Archive*, Report 2026/1344. **Preprint.**

[I.J.1] **Won Geun Shin**, JeongHwan Lee, Sangyun Jung, HeeSeok Kim. (2026). ARMOR: First-Order Masking of Activation and ArgMax Gadgets for Side-Channel Resistant Neural Networks. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, **2026**(2), 106-131. Published April 23, 2026. DOI: 10.46586/tches.v2026.i2.106-131 | Code

[I.C.1] MinAh Chae, **Won Geun Shin**, SangYun Jung, JiHun Yeom, DongHo Jeon, HeeSeok Kim. (2024). The Threat of Password Guessing Attacks Exploiting Linguistic Characteristics: A Case Study on the Korean Domains. *2024 Silicon Valley Cybersecurity Conference (SVCC)*, IEEE.

[D.J.1] JeongHwan Lee, Sang-Yun Jung, **Won Geun Shin**, Sujin Park, HeeSeok Kim. (2023). Trends in Side-Channel Countermeasures for LWE-like KEM and Digital Signature Schemes. *Review of KIISC*, 33(3), 85-96.

[D.C.1] **Won Geun Shin**, Seunghyeon Jeon, HeeSeok Kim. (2025). **Optimizing Side-Channel Leakage Assessment with Accumulated Distributed Algorithms.** *CISC-W'25.*

[D.C.2] **Won Geun Shin**, Sang-Yun Jung, HeeSeok Kim. (2023). **Side-Channel Countermeasure for Binary Neural Network Protection.** *CISC-S'23.*

HONORS AND AWARDS

KIISC President's Award - CISC 2025 *2025*

Best Poster Awards - IEEE SVCC 2024 *2024*

Signal Processing Research Institute President's Award - 9th Crypto Contest *2023*

KISA President's Award - CISC 2023 *2023*

777 Command Commander Award - 8th Crypto Contest *2022*

ETRI President's Award - Second Place in 5th Side-Channel Analysis Competition *2022*

PATENT

HeeSeok Kim, **Won Geun Shin**, SangYun Jung. (2024). **Side Channel Countermeasure for Activation Function of Binarized Neural Network.** Korean Patent No. 10-2024-0202916.

TALKS

Side-Channel Countermeasures and Security Evaluation. *Side Channel Analysis Tutorial (Workshop on Side Channel Analysis)*, July 1, 2025.

PROJECTS

A Study of Emerging Security Attack Techniques and Their Implementation on Commercial Chips *Oct 2025 - Present*
Side-Channel Attacks, Deep Learning

A Study on Physical Channel-based Hardware Vulnerability Validation of IoT Devices in a Black Box Environment *May 2024 - Present*
Fault Attacks, IoT Devices

Development of chip integrity and system security verification technology to ensure the safety of the HW supply chain *June 2024 - Present*
Side-Channel Attacks

Fault Injection Analysis of KpqC-Selected PQC Implementations *Apr 2025 - Oct 2025*
Fault Attacks, PQC

Empirical Study of Non-Invasive Countermeasures for Public-Key Cryptography on Commercial Platforms *Apr 2025 - Oct 2025*
Side-Channel Attacks

A Study on Security Vulnerabilities of Modern Processors Applicable to High-Risk Loss Environments *May 2024 - Oct 2024*
Side-Channel Attacks

Development of Physical Channel Vulnerability-based Attacks for Reliable On-Device Deep Learning Accelerator Design *Jan 2022 - Dec 2024*
Side-Channel Countermeasure, Side-Channel Attacks

Development of High-Performance Deep Learning-Based Side-Channel Analysis and Automation Techniques *Mar 2022 - Feb 2023*
Side-Channel Attacks, Deep Learning

A Study on Recent Trends in Fault Generation Mechanisms for Fault Injection Attacks *Apr 2022 - Oct 2022*
Fault Attacks